

Model-based control under observation noise

Domino recovers at 1 cm.

Boil succeeds with pose and reading noise.

Domino · 1 cm pose noise

100%

levels solved with MB features

Earlier MB: 50% · MF: 50%

Successful MB runs: 376.5 mean steps

Boil · pose + reading noise

100%

levels solved with MB features

MF: 25% · no matched MB-without run

Successful MB runs: 630 mean steps

Two seeds per noisy condition. All selected scorecards are final.

The noise channel is shared; the new belief tools are MB-only

Same observation channel

Gaussian noise on object pose and, for the combined boil point, scalar readings.

Sigmas are declared to both arms. Repeated observation at the same step returns the same draw.

MF stays a plain code agent and may write its own smoothing.

What we measure

Solve rate: fraction of train + test levels solved, over all seeds.

Steps: whole-run total, averaged only over seeds solving both levels.

Resets: mean across all seeds.

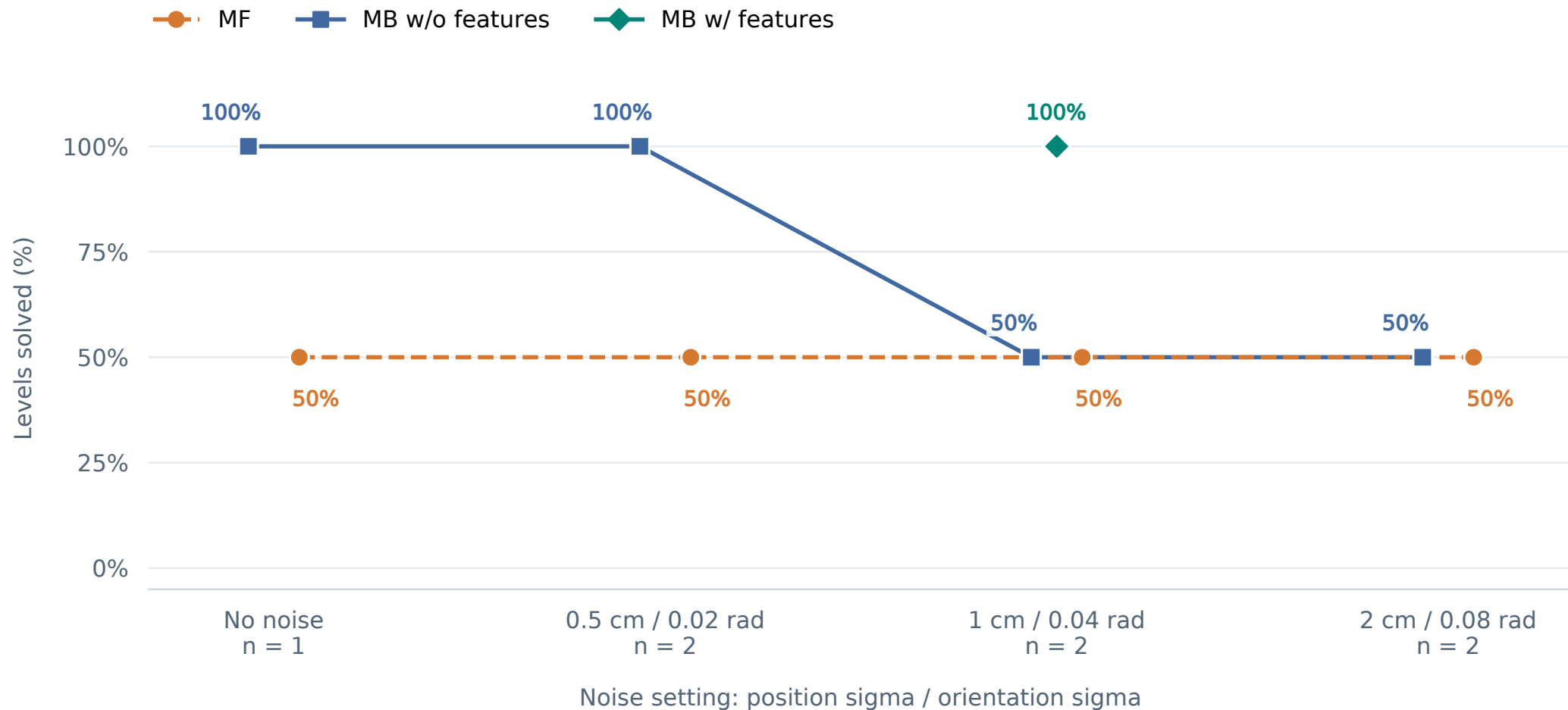
No successful seed means **N/A steps**, never zero steps.

Noiseless references: seed 0 only. Noisy conditions: seeds 0 and 1.

Skills and evaluators retain true state; this tests the agent's observation and decision channel.

Domino | Solve rate

Across observation-noise settings



Points = all-seed means. Noiseless: n = 1; noisy: n = 2. Solve rate includes train + test.

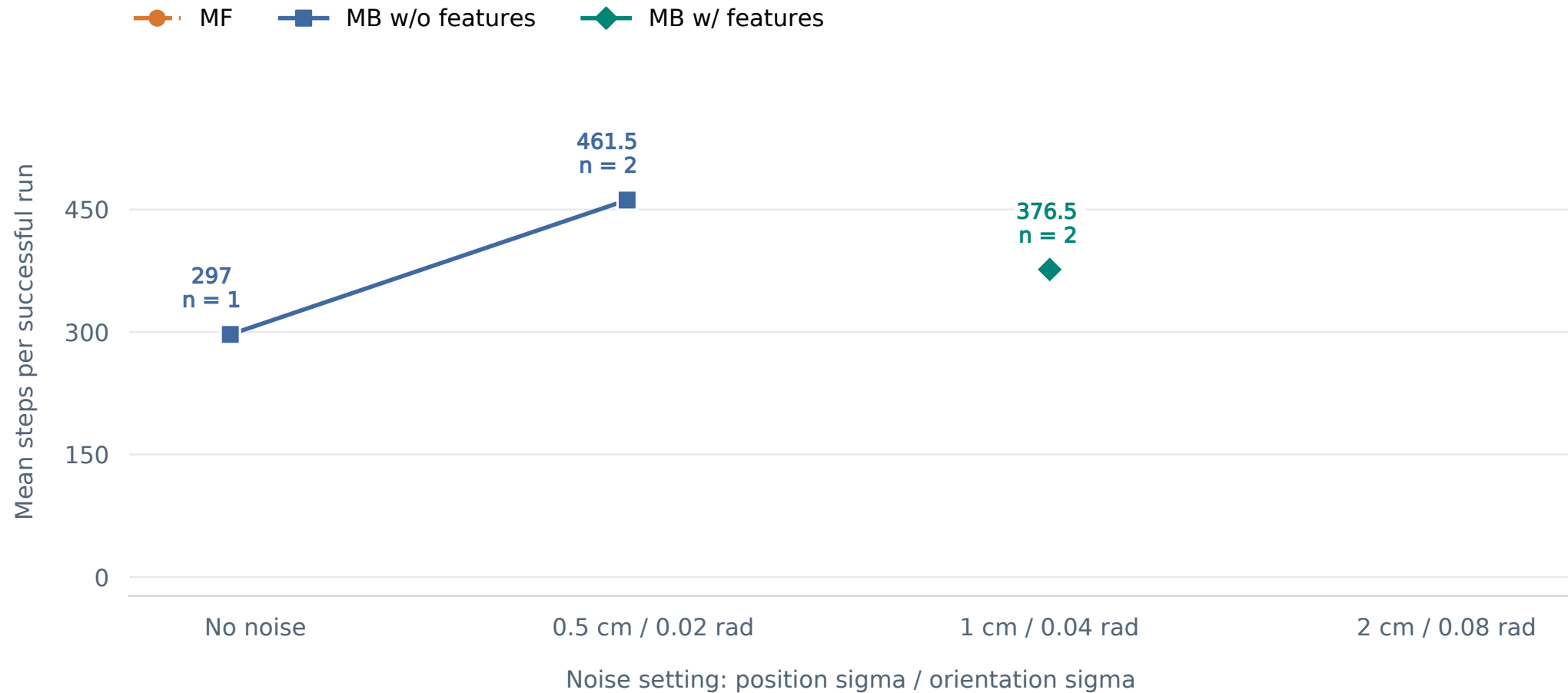
MB w/ features has one measured setting. Lines join measured adjacent settings; no extrapolation.

Earlier MB runs use older code. Read step costs alongside solve rate.

Categorical x-axis; small offsets reveal overlapping points. Snapshot: 2026-09-08 13:05 UTC.

Domino | Mean steps on successful seeds

Across observation-noise settings



Steps use only completed seeds solving both levels; n beside each point counts those seeds.

Missing step point: unmeasured setting or zero successful seeds. Other metrics use all seeds.

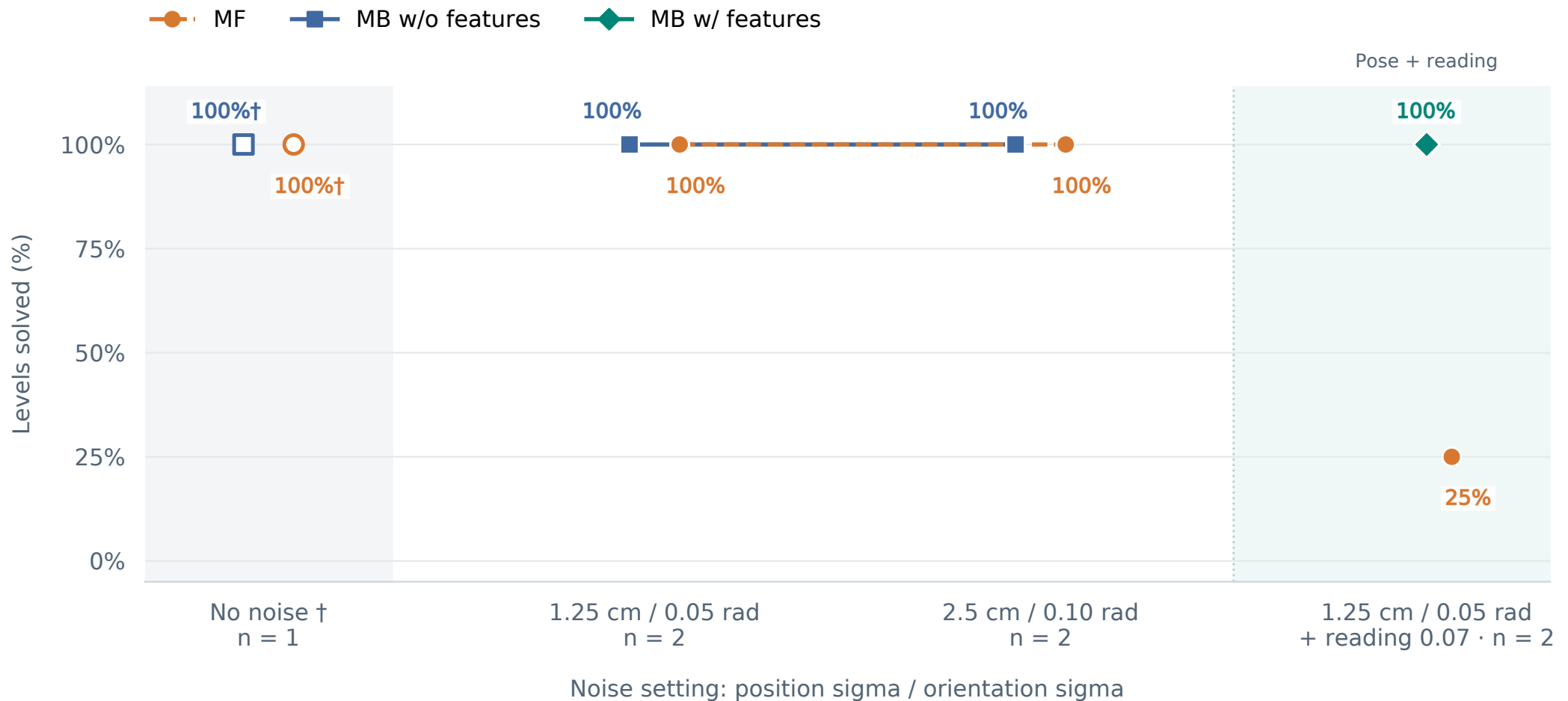
MB w/ features has one measured setting. Lines join measured adjacent settings; no extrapolation.

Earlier MB runs use older code. Read step costs alongside solve rate.

Categorical x-axis; small offsets reveal overlapping points. Snapshot: 2026-09-08 13:05 UTC.

Boil | Solve rate

Across observation-noise settings



Points = all-seed means. Noiseless: n = 1; noisy: n = 2. Solve rate includes train + test.

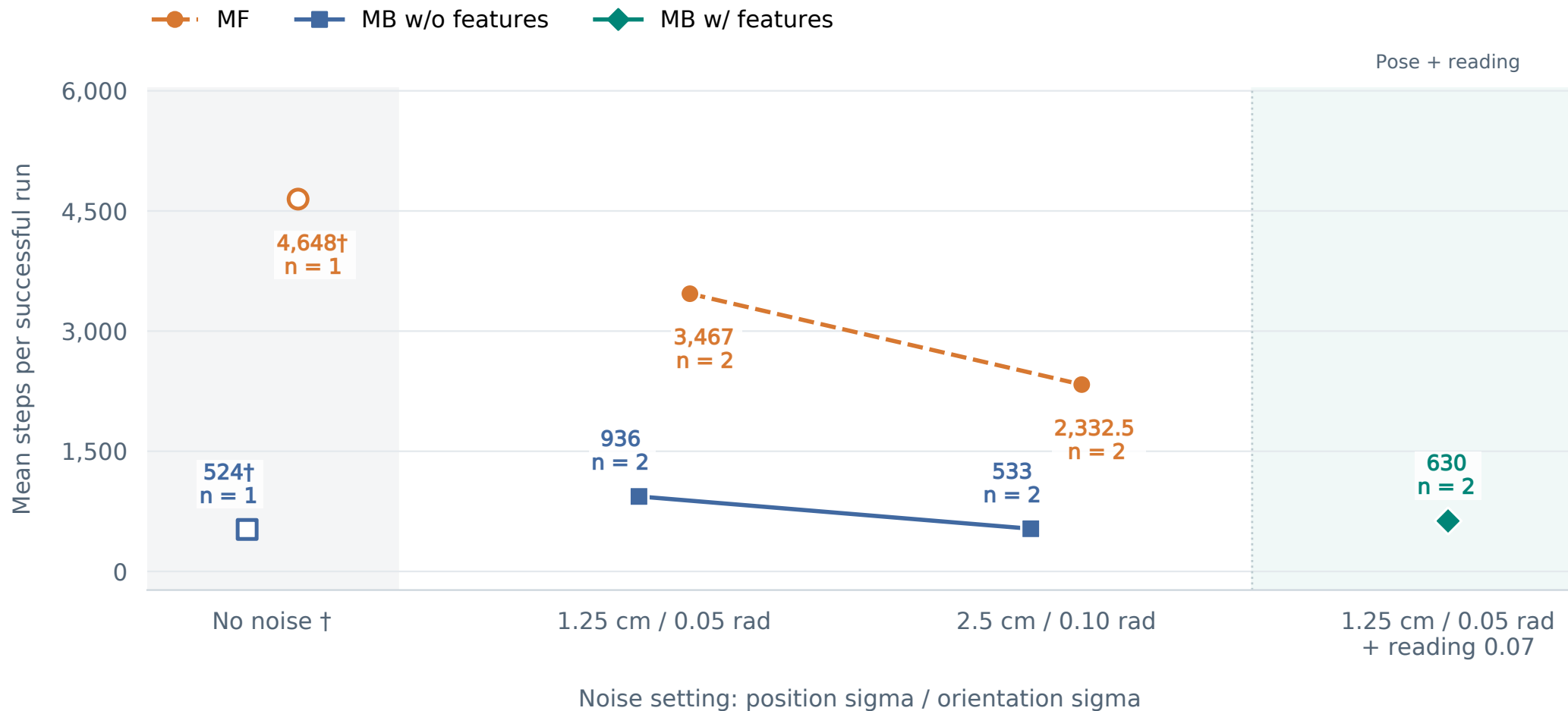
† Unsanitized historical references, shown separately. The combined point is not on the pose sweep.

MB w/ features has one measured setting. Read step costs alongside solve rate.

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Boil | Mean steps on successful seeds

Across observation-noise settings



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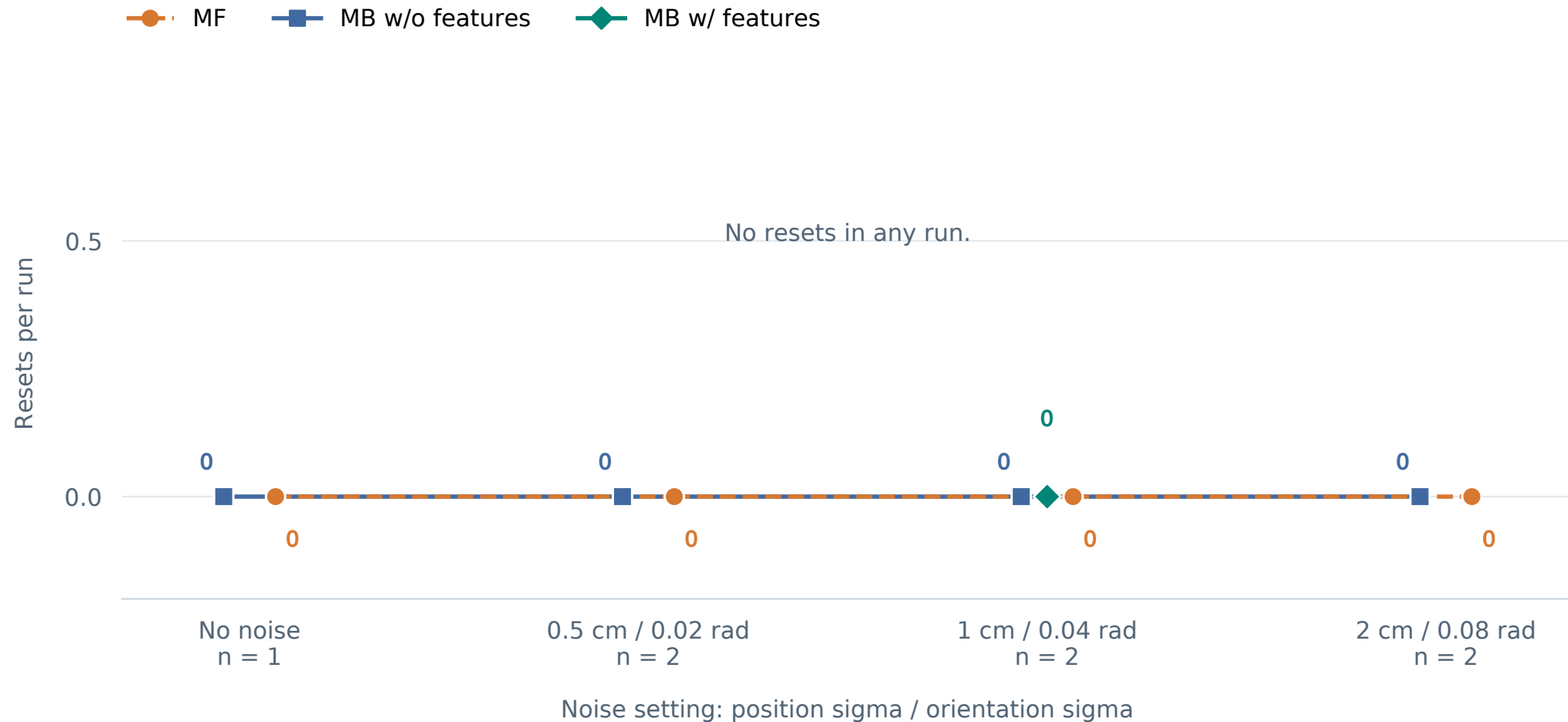
† Unsanitized historical references, shown separately. The combined point is not on the pose sweep.

MB w/ features has one measured setting. Read step costs alongside solve rate.

Categorical x-axis; small offsets reveal overlapping points. Snapshot: 2026-09-08 13:05 UTC.

Domino | Mean resets

Across observation-noise settings



Points = all-seed means. Noiseless: n = 1; noisy: n = 2. Solve rate includes train + test.

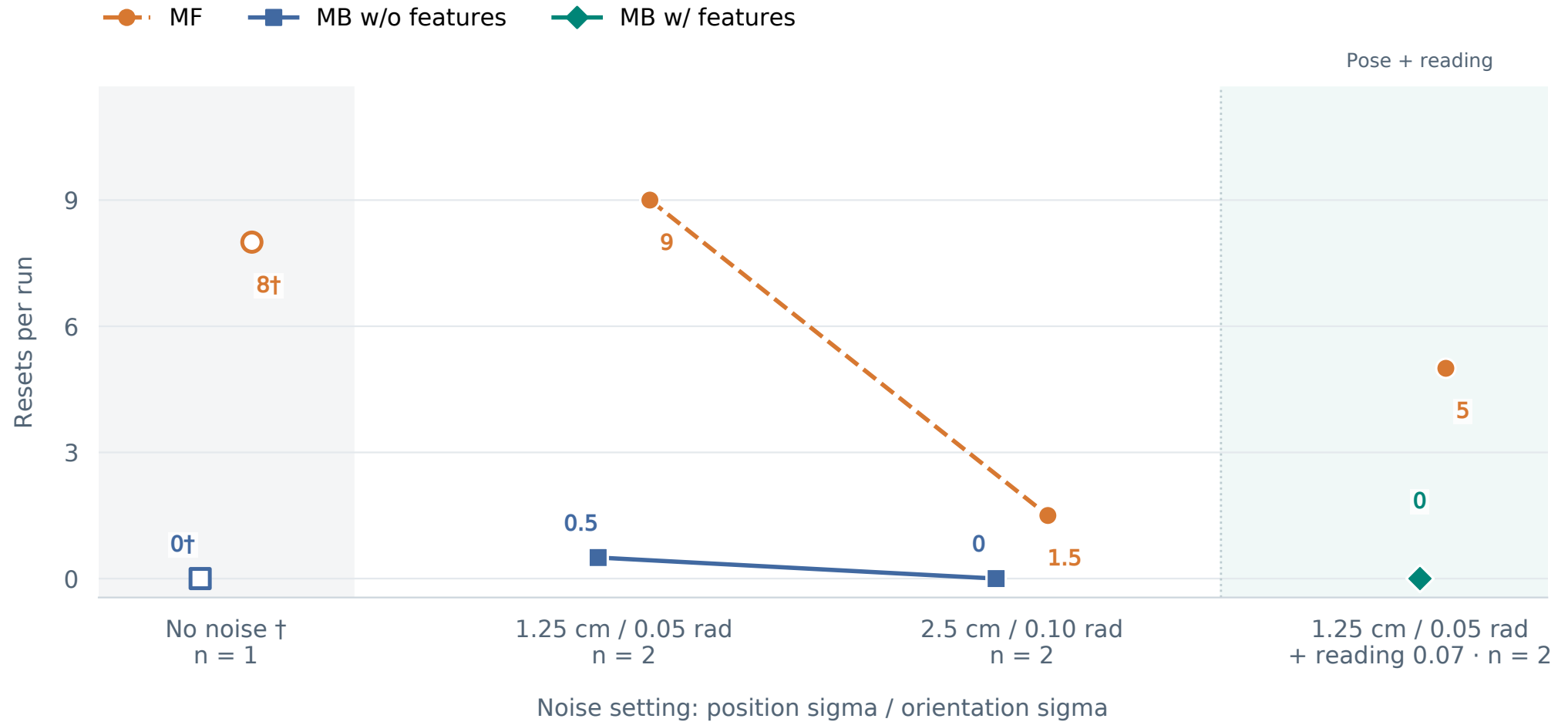
MB w/ features has one measured setting. Lines join measured adjacent settings; no extrapolation.

Earlier MB runs use older code. Read step costs alongside solve rate.

Categorical x-axis; small offsets reveal overlapping points. Snapshot: 2026-09-08 13:05 UTC.

Boil | Mean resets

Across observation-noise settings



Points = all-seed means. Noiseless: n = 1; noisy: n = 2. Solve rate includes train + test.

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MB w/ features has one measured setting. Read step costs alongside solve rate.

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Six flags, enabled together in the validation runs

01

Parameter intervals

Retain informative but wide fits and test plans across their intervals.

02

Noise-aware probes

Value a probe by whether its noisy observation can distinguish models.

03

Fit-side filtering

Find rest and motion relative to sigma; start rollouts from denoised rest windows.

04

Carried fit center

Use the last applied MAP as the next prior center; do not carry its width.

05

Laplace evidence

Report fit quality with a complexity penalty and comparable version deltas.

06

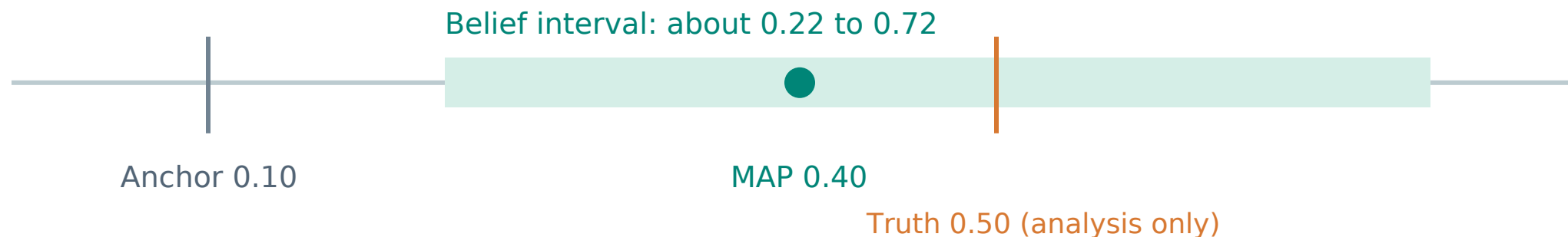
Execution belief

Expose smoothed state, spread and atom fractions for planning and monitoring.

The agent still writes the model and chooses actions.
The added machinery exposes uncertainty to those decisions.

A wide, informative fit should not fall back to an implausible anchor

Illustration from the earlier domino friction fit



Retain the estimate

A moved, informative but wide posterior gets the WIDE verdict and can deploy its MAP.

Test the interval

Plan sweeps report the fraction of parameter settings that pass and where outcomes change.

Probe when it matters

A mixed sweep can trigger information seeking. Probe value accounts for noisy readings.

Improve what the fit sees and what survives the next fit

Fit-side filtering

Detect motion from window means relative to the declared sigma.

Start a rest-anchored rollout from an averaged frame.

$$\text{spread} \approx \sigma / \sqrt{n}$$

This reduces initial-state noise while the object is at rest.

Carry the prior center

The last applied MAP becomes the next fit's prior center.

The width is not carried.

Fits already pool past levels' data; carrying its posterior width would count those data twice.

Compare by evidence

Report Laplace log evidence from the fit's residuals and local curvature.

Include a complexity penalty and comparable version deltas.

This informs the agent's choice; it is not an automatic model-replacement rule.

Expose spread, test sensitivity, then act

Execution-time belief

Average each object's noisy features over a recent window consistent with rest.

Shorten the window when its two half-means disagree beyond the noise threshold.

Expose the mean, spread and atom fractions beside the raw observation.

Slow motion can pass this heuristic and cause lag.
Expected-atom checks use fractions over belief draws.

```
sim.belief()
```

Read the current smoothed features, spreads and atom fractions.

```
sim.run(..., belief_draws=K)
```

Vary the starting state to test sensitivity to observation uncertainty.

```
sim.run(..., physics_sweep=True)
```

Sweep the learned parameter intervals.

```
evaluate_trajectory(...,  
physics_sweep=True)
```

Score the recorded sequence across parameter settings.

Not built: a full particle filter or observation-belief draws in `evaluate_trajectory`.
Atom fractions live in the observation and `sim.belief()`, not `sim.predicates()`.

Recovery is observed; the contribution of each feature is still open

What the runs show

Domino at 1 cm: both MB seeds now solve both levels; the earlier MB pair did not.

Boil, combined noise: MB solves all four levels across seeds, with zero resets; MF solves one.

All four new MB runs finish successfully.

What they do not isolate

Only two noisy seeds and one noiseless reference seed per arm.

Earlier MB runs use older code; individual feature ablations are still absent.

Boil has no matched feature-off run with reading noise, and its old noiseless references exposed hidden heat.

To support stronger claims: clean noiseless boil controls, matched feature ablations, and more seeds.

The six MB feature flags

Feature	Flag enabled in the validation MB configs
Parameter intervals	<code>code_sim_learning_interval_belief</code>
Noise-aware probes	<code>agent_explorer_info Seeking_noise_aware</code>
Fit-side filtering	<code>code_sim_learning_rollout_noise_filter</code>
Carried fit center	<code>code_sim_learning_carry_posterior</code>
Laplace evidence	<code>code_sim_learning_fit_evidence</code>
Execution belief	<code>continual_belief_frame</code>

All six default off.
Shared channel: `continual_obs_noise_position`, `continual_obs_noise_orientation` and `continual_obs_noise_scalar`.

Seed means and successful-run counts

Env	Noise σ	Arm	n	Success n	Solve	Steps ¹	Resets
Domino	No noise	MB	1	1	100%	297	0
Domino	No noise	MF	1	0	50%	N/A	0
Domino	0.5 cm/0.02 rad	MB	2	2	100%	461.5	0
Domino	0.5 cm/0.02 rad	MF	2	0	50%	N/A	0
Domino	1 cm/0.04 rad	MB	2	0	50%	N/A	0
Domino	1 cm/0.04 rad	MF	2	0	50%	N/A	0
Domino	1 cm/0.04 rad	MB + features	2	2	100%	376.5	0
Domino	2 cm/0.08 rad	MB	2	0	50%	N/A	0
Domino	2 cm/0.08 rad	MF	2	0	50%	N/A	0
Boil	No noise	MB †	1	1	100%	524	0
Boil	No noise	MF †	1	1	100%	4,648	8
Boil	1.25 cm/0.05 rad	MB	2	2	100%	936	0.5
Boil	1.25 cm/0.05 rad	MF	2	2	100%	3,467	9
Boil	2.5 cm/0.10 rad	MB	2	2	100%	533	0
Boil	2.5 cm/0.10 rad	MF	2	2	100%	2,332.5	1.5
Boil	1.25 cm/0.05 rad + read 0.07	MB + features	2	2	100%	630	0
Boil	1.25 cm/0.05 rad + read 0.07	MF	2	0	25%	N/A	5

¹ Steps use only seeds solving both levels; N/A = zero successful seeds. Solve rate and resets use all seeds.

† Unsanitized historical boil reference. Position σ is in cm; orientation σ is in radians.

Source: uncertainty-results/snapshot.json, captured 2026-09-08T13:05:10.283352+00:00; 17 conditions / 30 runs.